

Eenheid 1.2

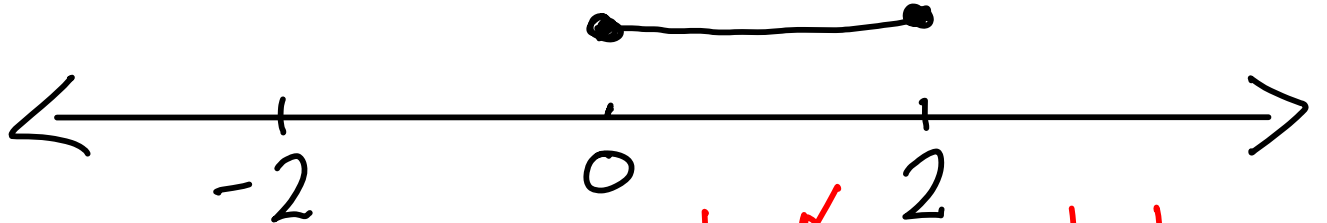
Note Title

2017/02/06

Die Absolute Waarde

Stewart, Appendix A p A6 - A8
(zie in handboek)

Meetkundige Betekenis : $|2|$

Getallenlijn: 

- $|2|$: afstand van 2 na nul: $|2 - 0| = |2| = 2$
- $|-2|$: afstand van -2 na nul: $|(-2) - 0| = |-2| = 2$
- Afstand is positief

Definisie (Memoriseer Definities)

Absolute Waarde : p A6

Laat $x \in \mathbb{R}$, dan is $|x|$ die afstand van x na 0 op die getalreël. $(|x - 0| = |x|)$

$$|x| = \begin{cases} x & \text{as } x \geq 0 \\ -(x) & \text{as } x < 0 \end{cases}$$

✓ of nul
punte

Antwoord is altyd positief.

Bv. $|-3| = 3$

$$|2.345| = 2.345$$

Eienskappe : $p \neq 7$

Absolute waarde

Laat $a, b \in \mathbb{R}$ en $n \in \mathbb{Z}$, dan is

1. $|ab| = |a||b|$

2. $\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$, $b \neq 0$

3. $|a^n| = |a|^n$

Voeg by:

• 4. $\sqrt{a^2} = |a|$

• 5. $|a| \geq 0$ vir alle $a \in \mathbb{R}$

• 6. $|a| = 0 \Leftrightarrow a = 0$

• 7. $|a| \leq |b| \Leftrightarrow a^2 \leq b^2$
 $|a| < |b| \Leftrightarrow a^2 < b^2$ } *Sien Studiegids*

" $\Leftrightarrow$ " : "as en steps as"

• Voorbeeld: $|x-2| \leq |x+1|$
..... (Toepassing van
eigenschap 7)

$\Rightarrow \sqrt{(x-2)^2} \leq \sqrt{(x+1)^2}$ eigenschap 4)
kwadrateer beide kanten

$\Rightarrow (x-2)^2 \leq (x+1)^2$ eigenschap 7

$\Rightarrow x^2 - 4x + 4 \leq x^2 + 2x + 1$

$\Rightarrow -4x - 2x \leq 1 - 4$

$\Rightarrow 6x \geq 3$

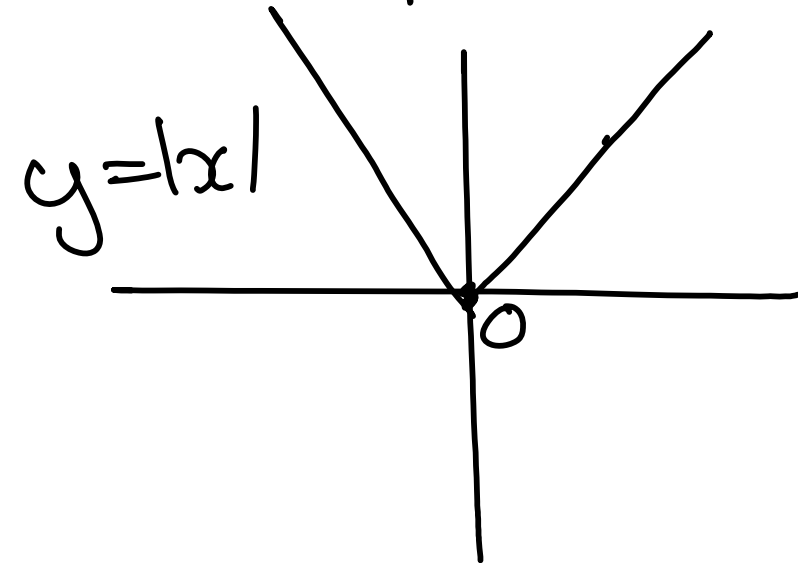
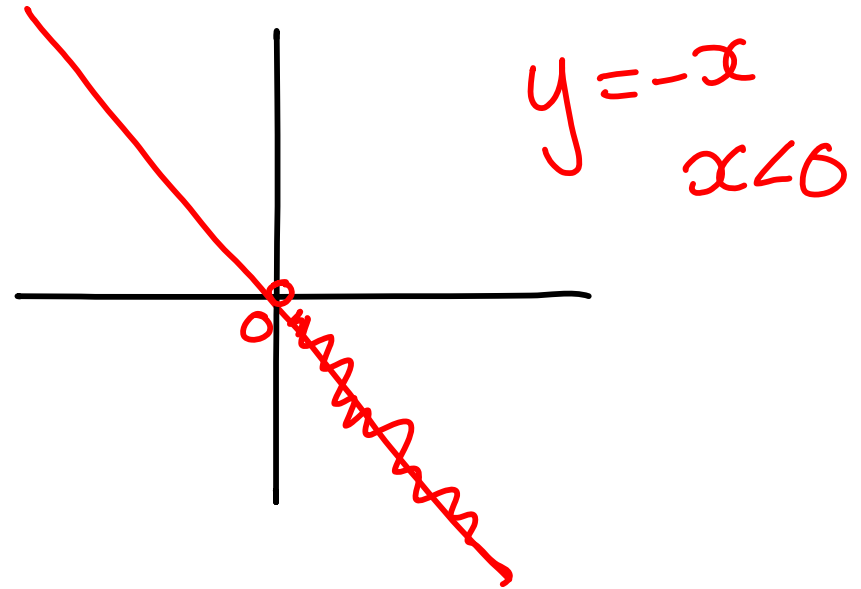
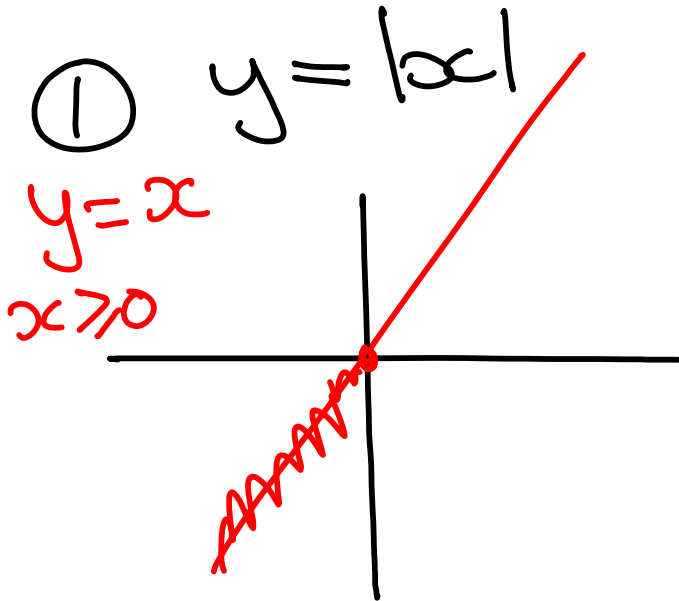
$\Rightarrow x \geq \frac{1}{2}$

$\Rightarrow x \in [\frac{1}{2}, \infty)$

Sketse van Absolute waarde:

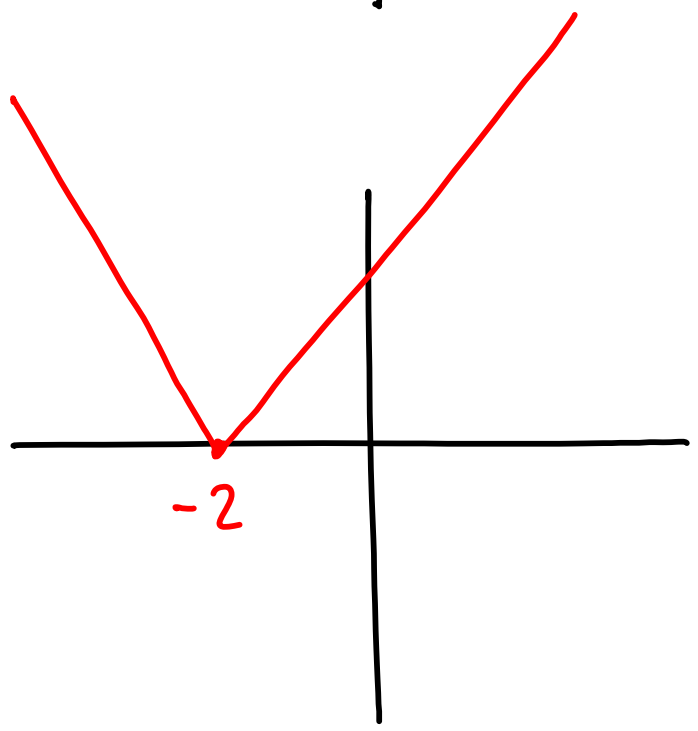
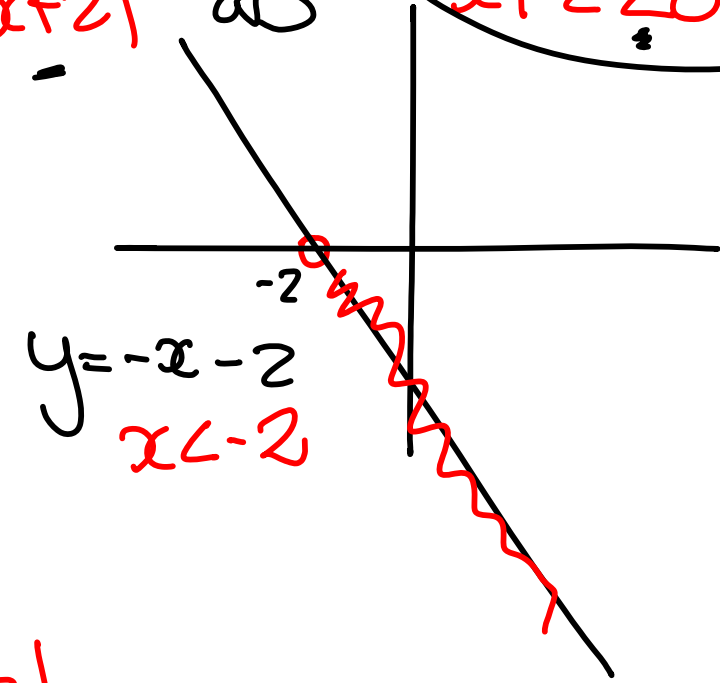
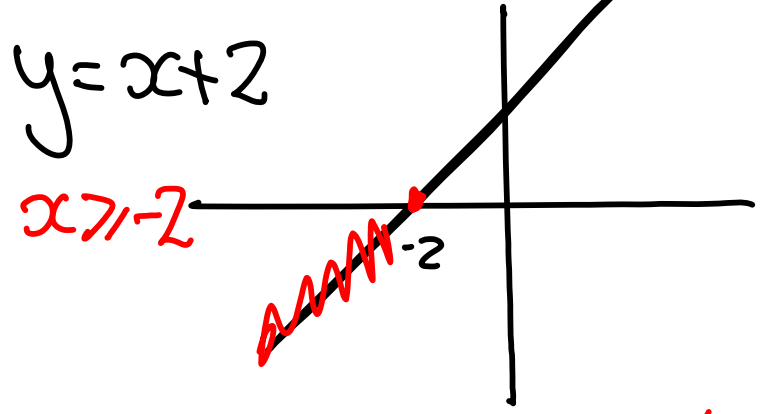
Gebruik Definitie: $|x| = \begin{cases} x & \text{as } x \geq 0 \\ -x & \text{as } x < 0 \end{cases}$

Definering



② $y = |x+2| = \begin{cases} x+2 & \text{as } x+2 \geq 0 \\ -(x+2) & \text{as } x+2 < 0 \end{cases}$

$x+2 \geq 0$
 $x+2 < 0$



$y = |x + 2|$

③ $y = f(x) = |x^2 - 4|$ Stuksgewijs gedefinieerde functie

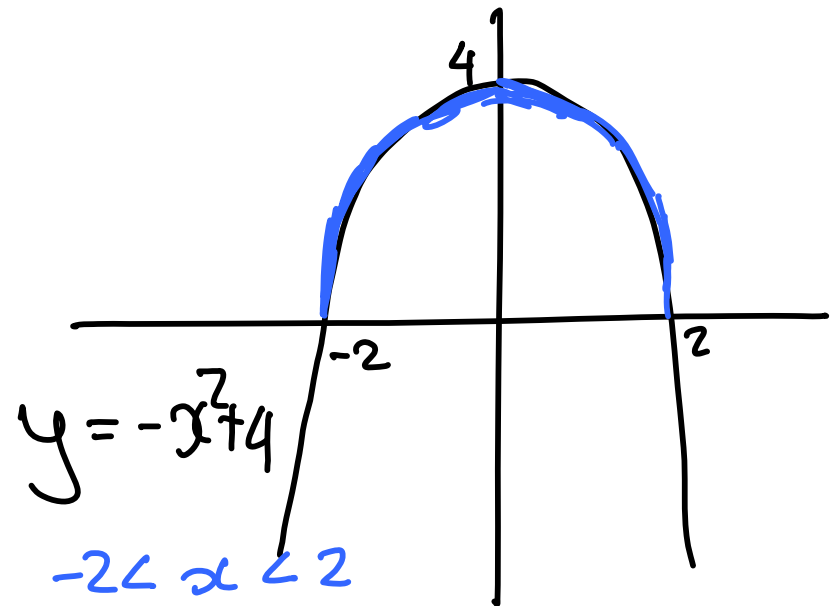
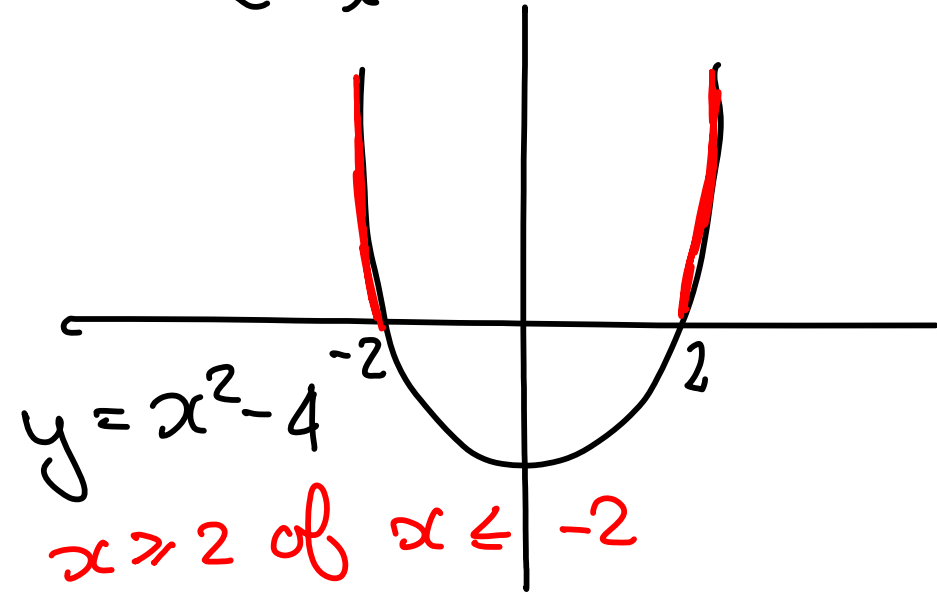
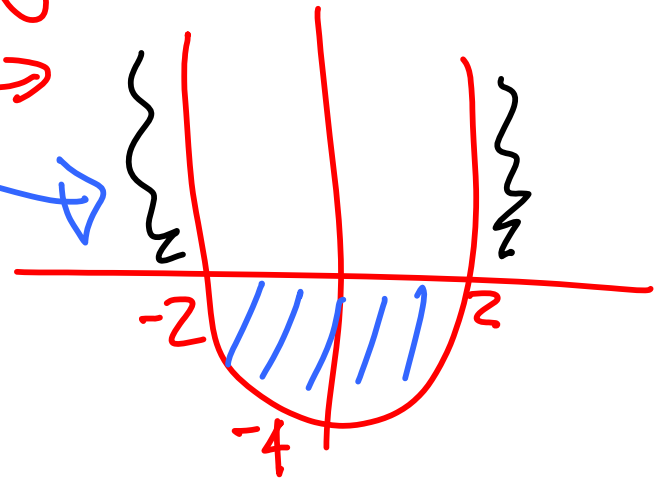
Bepaling

$$x^2 - 4 \geq 0$$

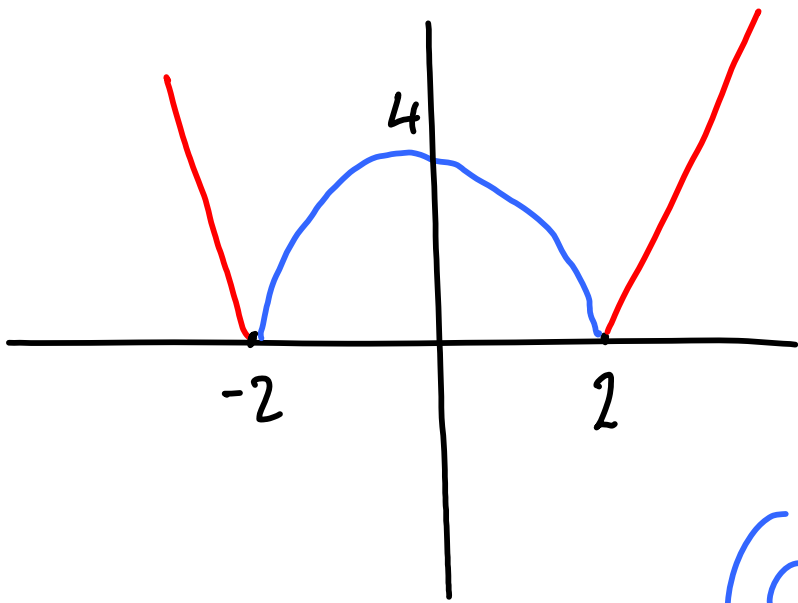
$$x^2 - 4 < 0$$

$$= \begin{cases} x^2 - 4 & \text{as} \\ -(x^2 - 4) & \text{as} \end{cases}$$

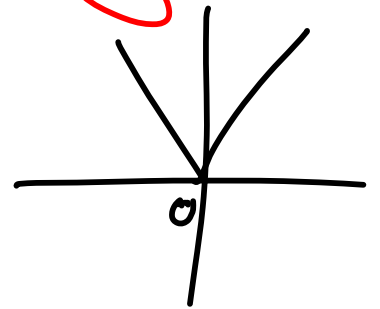
$$= \begin{cases} x^2 - 4 & \text{as } x \geq 2 \text{ of } x \leq -2 \\ -x^2 + 4 & \text{as } -2 < x < 2 \end{cases}$$



$$y = |x^2 - 4| = \begin{cases} x^2 - 4 & \text{as } x \geq 2 \text{ of } x \leq -2 \\ -x^2 + 4 & \text{as } -2 < x < 2 \end{cases}$$



• Definitiesverzameling van $y = |x|$ (Gebied)



$$D_f = \mathbb{R} = \{x \mid x \in \mathbb{R}\}$$

$$\bullet y = |x| \geq 0$$

$$D_f = \mathbb{R}$$

$$\bullet y = |x| > 0$$

$$D_f = \{x \mid x \neq 0\} = \mathbb{R} \setminus \{0\}$$

Los gelykhede en ongelijkheden op
met absolute waarden.

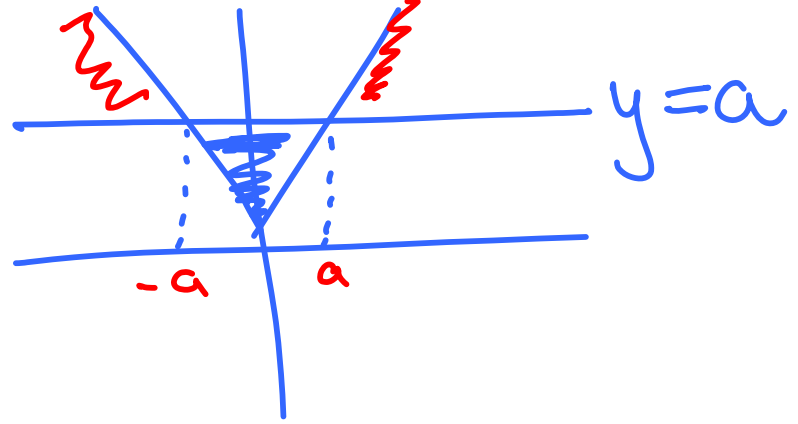
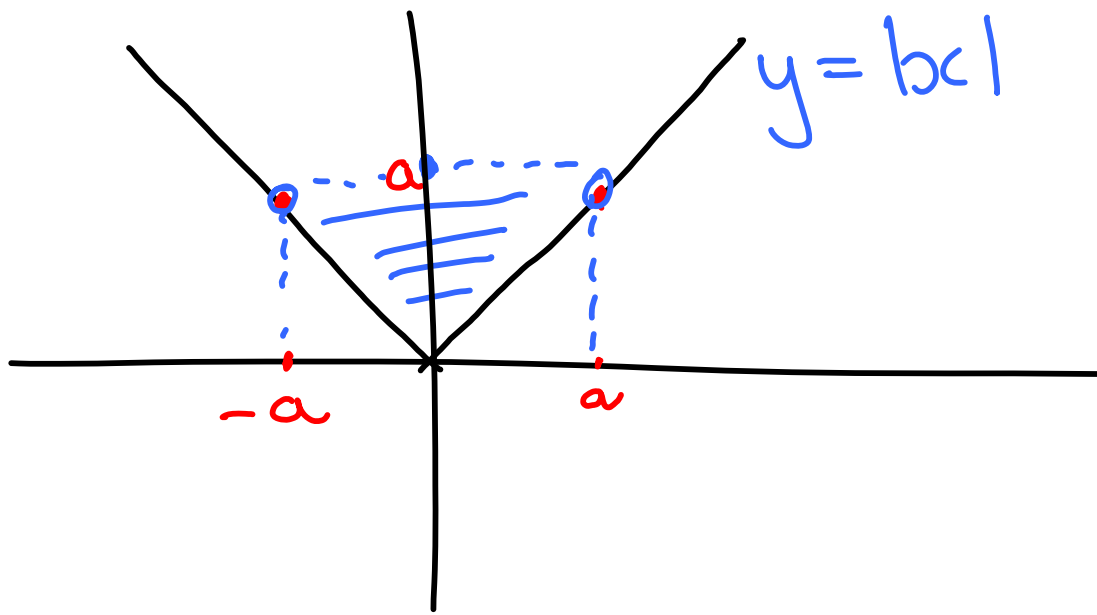
Stelling [6] $q \equiv p \wedge$

Laat $y = |x|$, $a \in \mathbb{R}$, $a > 0$

✓ $y = |x| = a$ is en slechts is $x = \pm a$
 $\Leftrightarrow x = a$ of $x = -a$

$\equiv$ $y = |x| < a \Leftrightarrow -a < x < a$

$\Leftrightarrow y = |x| > a \Leftrightarrow x > a$ of $x < -a$



Gebruik die stelling en los op:

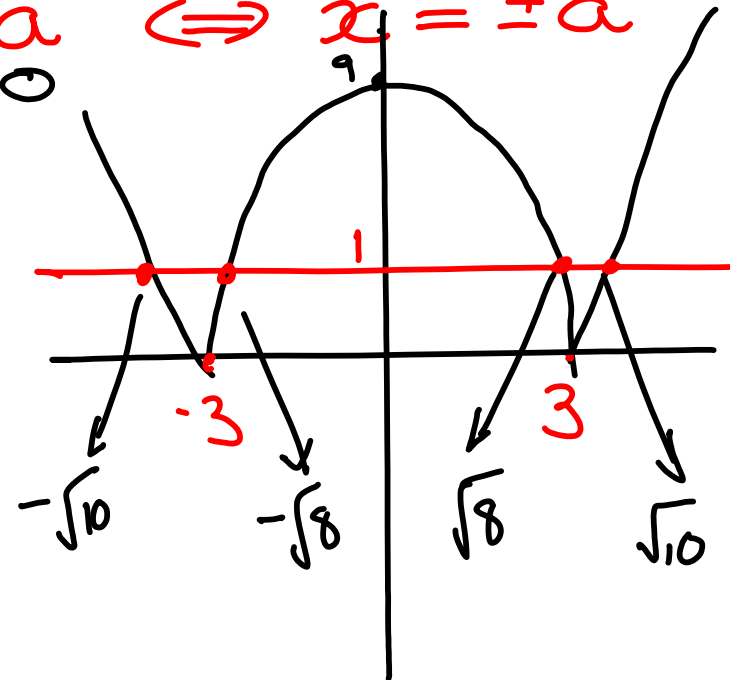
① $|x^2 - 9| = 1$

$\therefore x^2 - 9 = 1$ of $x^2 - 9 = -1$

$\Rightarrow x^2 = 10$ of $x^2 = 8$

$\Rightarrow x = \pm\sqrt{10}$ of $x = \pm\sqrt{8}$

* $|x| = a \Leftrightarrow x = \pm a$
 $a > 0$



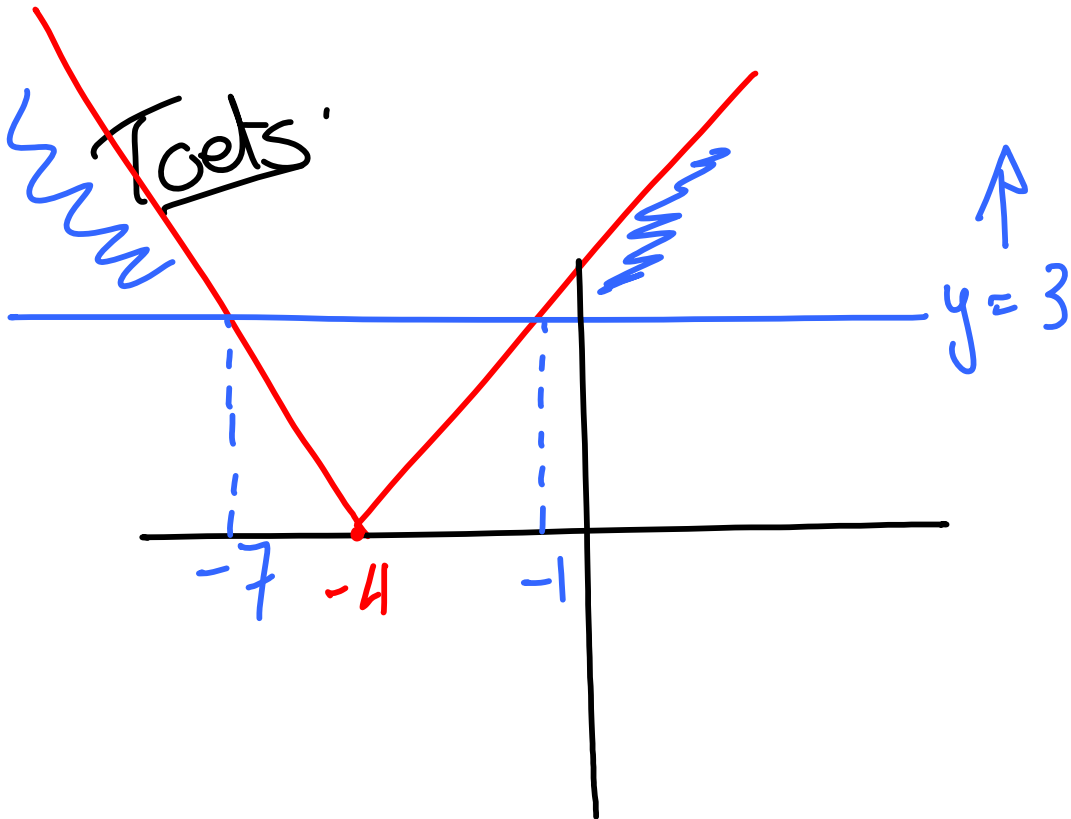
$$\textcircled{2} \quad |x+4| > 3$$

$$\Rightarrow x+4 > 3 \text{ of } x+4 < -3$$

$$\Rightarrow x > -1 \text{ of } x < -7$$

$$\Rightarrow x \in (-\infty, -7) \cup (-1, \infty)$$

Onthou Stelling
 $a > 0$
 $|x| > a$
 $\Leftrightarrow x > a \text{ of } x < -a$



$$(3) \quad |1-x| \leq 4$$

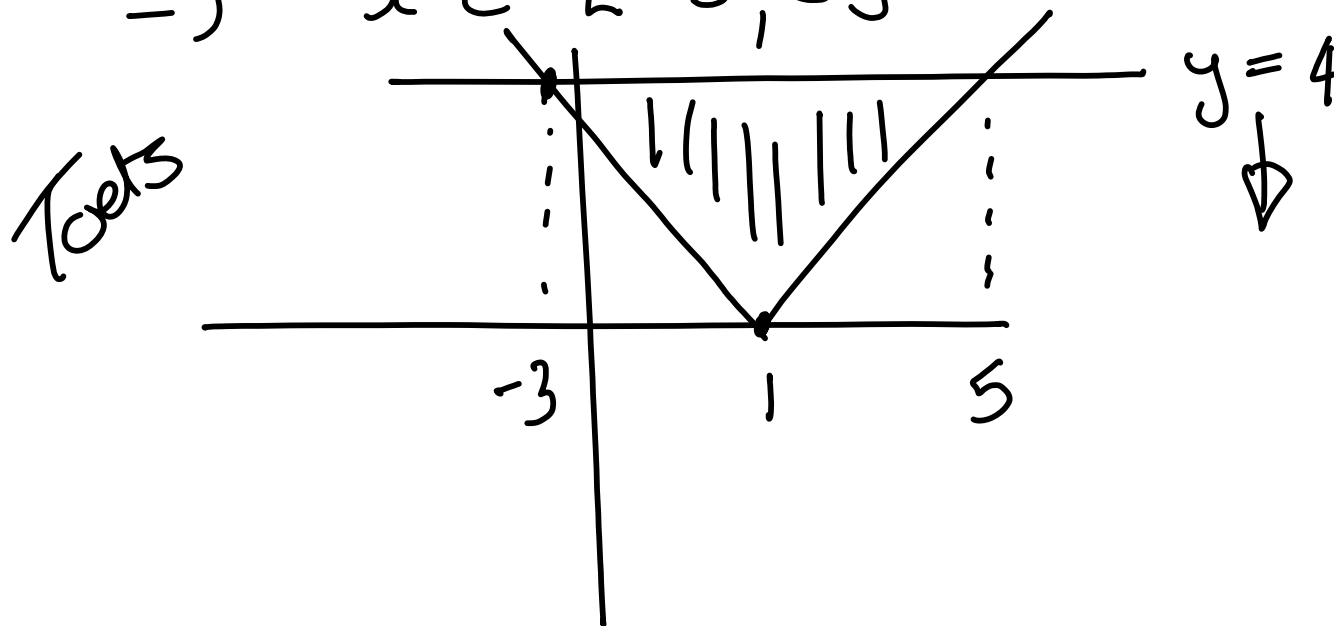
$$\Rightarrow -4 \leq 1-x \leq 4$$

$$\Rightarrow -4-1 \leq -x \leq 4-1$$

$$\Rightarrow -5 \leq -x \leq 3$$

$$\Rightarrow 5 \geq x \geq -3$$

$$\Rightarrow x \in [-3, 5]$$



Stelling: $a > 0$

$$|x| \leq a \Leftrightarrow$$

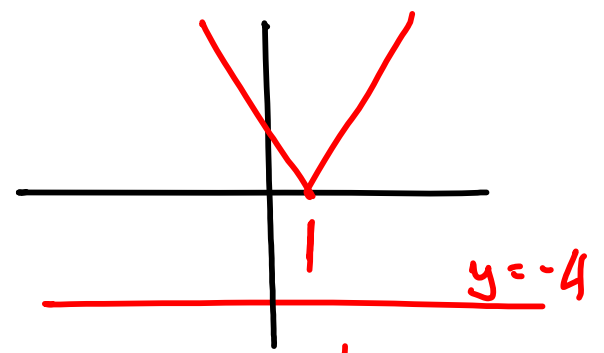
$$-a \leq x \leq a$$

$$\Rightarrow -3 \leq x \leq 5$$

$$D_f = [-3, 5]$$

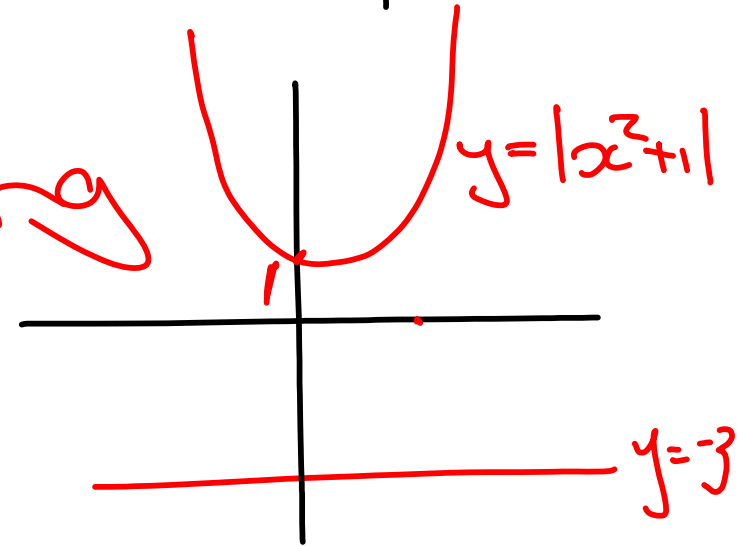
$$= \{x \mid -3 \leq x \leq 5\}$$

④ $|x-1| < -4$ (Inspeksie)
 $\Rightarrow x \in \emptyset$



⑤ $|x^2+1| > -3$ (Inspeksie)
 $\Rightarrow x \in \mathbb{R}$

Regstelling



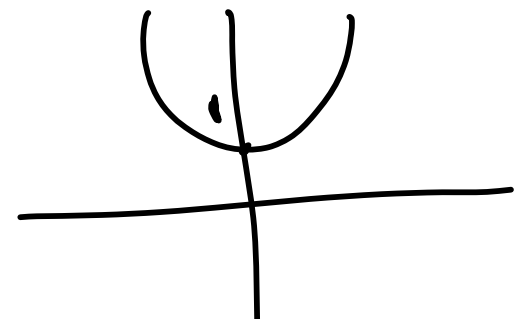
Definisie van $|x^2+1|$:

$$|x^2+1| = \begin{cases} x^2+1 & \text{as } x^2+1 \geq 0 \\ -(x^2+1) & \text{as } x^2+1 < 0 \end{cases}$$

???

$\rightarrow x^2 < -1$??

$= x^2+1$ as $x^2+1 \geq 0$



Opmerking:

Die afstand tussen x en y op die reële getalrelyn is $|x-y|$.

Die afstand tussen -5 en -7 is:

$$|-5 - (-7)| = |-5 + 7| = |2| \quad \text{Of}$$

$$|-7 - (-5)| = |-2| = 2$$

Los op: ① $\frac{|x-1|^{(+)}}{2-x?} > 0$ ant (+)

Onthou: $x \neq 2$

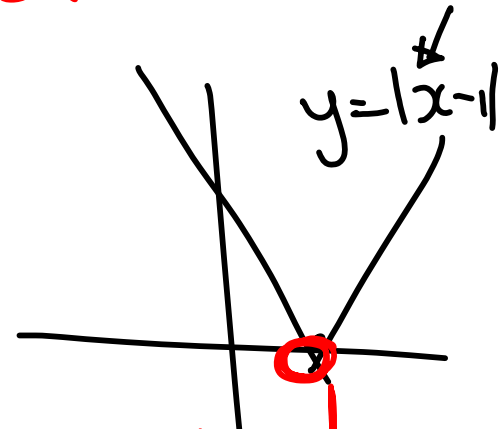
$$\Rightarrow |x-1| > 0 \text{ en } 2-x > 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (1, \infty) \text{ en } x < 2$$

$x \neq 1$



$$\Rightarrow x \in (-\infty, 1) \cup (1, 2)$$

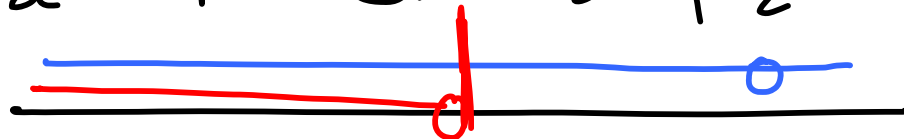


$$Df = \mathbb{R} \setminus \{1, 2\}$$

$$\textcircled{2} \quad \frac{x-1}{|2-x|} < 0$$

$$\Rightarrow x-1 < 0 \text{ en } |2-x| > 0$$

$$\Rightarrow x < 1 \text{ en } x \neq 2$$

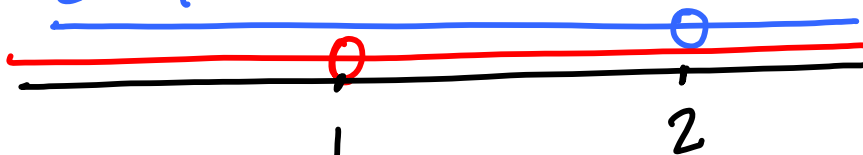


$$\Rightarrow x < 1 \Rightarrow x \in (-\infty, 1)$$

$$\textcircled{3} \quad \frac{|x-1|}{|2-x|} > 0$$

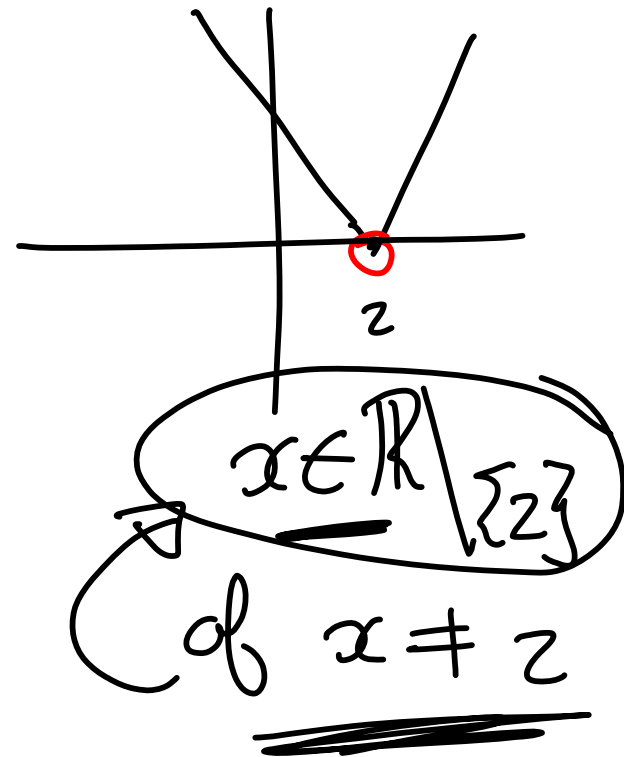
$$\Rightarrow |x-1| > 0 \text{ en } |2-x| > 0$$

$$\Rightarrow x \neq 1 \text{ en } x \neq 2$$



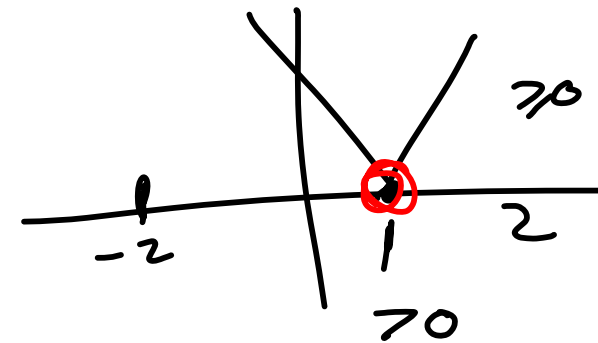
$$\Rightarrow x \in \mathbb{R} \setminus \{1, 2\}$$

$$\Rightarrow x \in (-\infty, 1) \cup (1, 2) \cup (2, \infty)$$



$$\text{of } x \in \mathbb{R} \setminus \{1\}$$

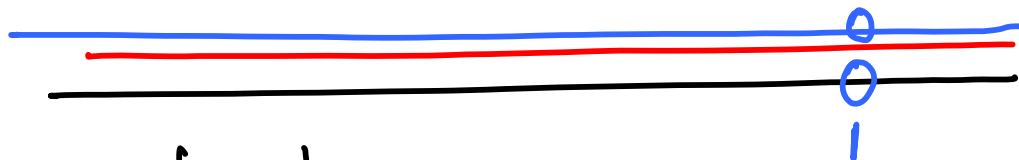
$$x \in \mathbb{R} \setminus \{2\}$$



$$\textcircled{4} \quad \frac{|x|}{|x-1|} \geq 0$$

$$\Rightarrow |x| \geq 0 \text{ en } |x-1| > 0$$

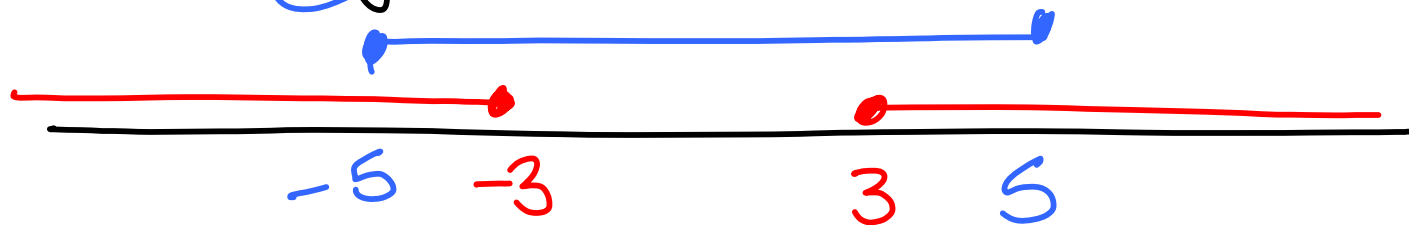
$$\Rightarrow x \in \mathbb{R} \text{ en } x \neq 1 \Rightarrow x \in (-\infty, 1) \cup (1, \infty)$$



$$\textcircled{5} \quad 3 \leq |x| \leq 5 \quad (\text{stelling})$$

$$\Rightarrow |x| \geq 3 \text{ en } |x| \leq 5$$

$$\Rightarrow x \geq 3 \text{ of } x \leq -3 \text{ en } -5 \leq x \leq 5$$



$$\Rightarrow x \in [-5, -3] \cup [3, 5]$$

3

Interessante voorbeeld

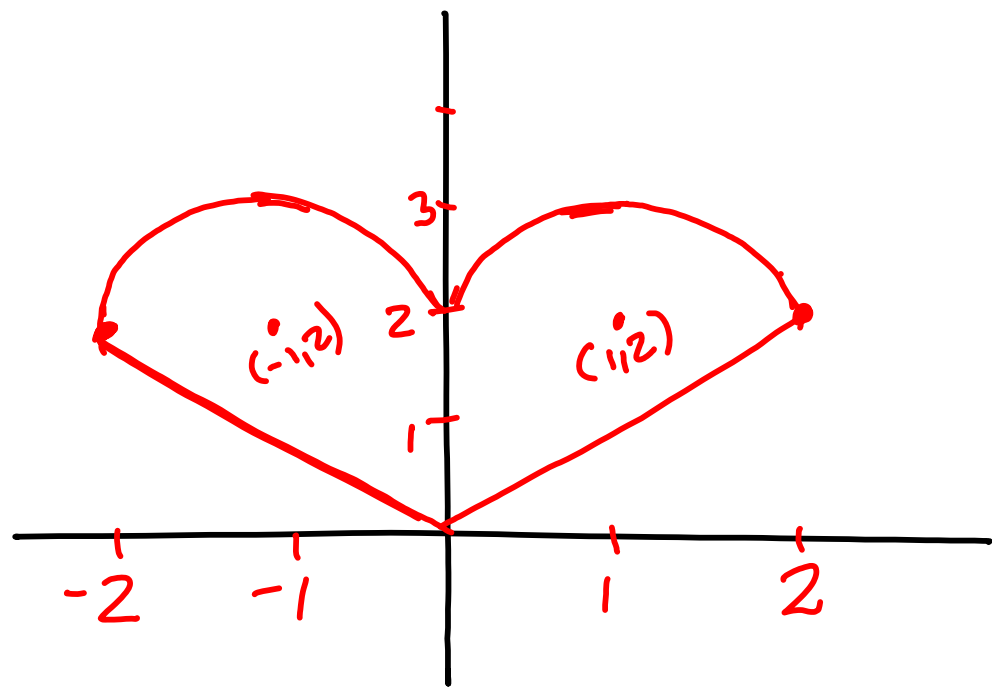
Skets die volgende op een assenstelsel:

$$\left. \begin{array}{l} 1) \ (x-1)^2 + (y-2)^2 = 1, \ y \geq 2 \\ (x+1)^2 + (y-2)^2 = 1, \ y \geq 2 \\ y = |x|, \ -2 \leq x \leq 2 \end{array} \right\}$$

$$2) \ g(x) = \frac{3x + |x|}{x}$$

Oplossing: 1)

Gelukkige
Valentynsdag
vir jou ♥



$$2) \quad g(x) = \frac{3x + |x|}{x} = \begin{cases} \frac{3x + x}{x} & \text{as } x > 0 \\ \frac{3x + (-x)}{x} & \text{as } x < 0 \end{cases}$$

$$= \begin{cases} \frac{4x}{x} & \text{as } x > 0 \\ \frac{2x}{x} & \text{as } x < 0 \end{cases}$$

